



Pyrolysis of butane-2,3-dione from low to high pressures: Implications for methyl-related growth chemistry

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ABSTRACT

In order to investigate the pressure-dependent effects of carbon chain growth from methyl radical, pyrolysis experiments of butane-2,3-dione (also known as diacetyl) were conducted both in a flow reactor over 780–1520 K at low and atmospheric pressures and in a jet-stirred reactor (JSR) over 650–1130 K at 10 bar. Identification and quantification of the intermediates were achieved by synchrotron vacuum ultraviolet photoionization mass spectrometry in a flow reactor and Fourier transform infrared spectrometry and gas chromatography in a JSR. A pyrolysis model of butane-2,3-dione, incorporating the sub-mechanisms of butane-2,3-dione, ketene, C1–C4 hydrocarbons and the formation of benzene, was developed from our recently reported C1 model in this work. It was validated against the present pyrolysis data of butane-2,3-dione and the pyrolysis data of C2–C4 hydrocarbon fuels and ketene in the literature. Generally, the present model could reasonably predict all the experimental targets. Based on the rate of production analysis, the unimolecular decomposition reaction ($\text{CH}_3\text{COCOCH}_3 (+\text{M}) = 2\text{CH}_3\text{CO} (+\text{M})$) is the dominant pathway to consume the fuel at low pressure. However, at atmospheric and high pressures, both the unimolecular decomposition reaction and the H abstraction reaction of butane-2,3-dione by methyl radical are important. Ketene, acetone and acetaldehyde were measured in this work and their formation is also pressure dependent. For the methyl-related growth chemistry, addition-elimination reactions play an important role at low pressure. Their contribution becomes less at atmospheric pressure and can be neglected at high pressure. In contrast, the combination reactions are responsible for methyl-related growth at high pressure. For the formation of benzene, the recombination of propargyl radical is the most important pathway at low pressure, while the H-assisted isomerization reaction of fulvene becomes dominant at atmospheric and high pressures.

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1. Introduction

The kinetics of methyl radical (CH_3) not only plays a crucial role in the core hydrocarbon oxidation mechanism [1], but is also critical to predict global combustion parameters of large hydrocarbon and oxygenated fuels, such as ignition delay times [2] and laminar flame speeds [3]. Under fuel-rich combustion of methane [4], CH_3 -related combination and addition-elimination reactions (e.g.

R1 and R2, respectively) are responsible for the growth of carbon chain and the formation of soot precursors. Furthermore, the H-abstraction- CH_3 -addition mechanism was recently highlighted by Hansen et al. [5] for the formation and ring expansion of aromatic hydrocarbons. Therefore, a better understanding of the CH_3 kinetics will contribute to accurate predictions of both global combustion parameters and pollutant formation.



Due to the critical role of CH_3 kinetics, numerous studies have been reported on the CH_3 -related elementary reactions, some of which are oxygen-absent reactions, such as the H-abstraction

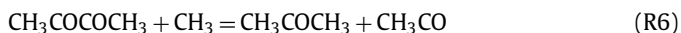
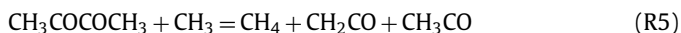
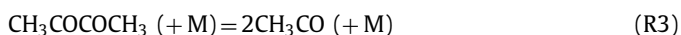
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reactions by CH_3 [6,7], combination reactions [8–12] (e.g. R1) and addition-elimination reactions (e.g. R2) [7,13]. Previous fundamental experiments under fuel-rich conditions, aiming at providing various validation targets for the CH_3 -related growth chemistry, were usually designed and conducted with methane [4,14,15]. CH_3 could be produced from methane under pyrolysis circumstance, which becomes inefficient below 1500 K [15]. Therefore, in order to provide validation data for the CH_3 -related growth chemistry at temperatures below 1500 K, especially under the intermediate temperature conditions, more efficient source to generate CH_3 is needed.

Butane-2,3-dione ($\text{CH}_3\text{COCOCH}_3$, also known as diacetyl) is able to effectively generate CH_3 under pyrolysis circumstance at temperatures below 1500 K compared with other precursors like methane, ethane, acetaldehyde (CH_3CHO) and acetone (CH_3COCH_3) [10]. According to previous studies on butane-2,3-dione pyrolysis [16–23], it is initiated by C–C bond scission (R3) to produce two acetyl radicals (CH_3CO), which rapidly decompose to CH_3 and CO (R4). The formed CH_3 abstracts an H atom from $\text{CH}_3\text{COCOCH}_3$ to generate $\text{CH}_3\text{COCOCH}_2$, which quickly dissociates to yield ketene (CH_2CO) and CH_3CO (R5). Besides, CH_3COCH_3 is also an important product of butane-2,3-dione pyrolysis [20], and mainly produced through an addition-elimination reaction between CH_3 and butane-2,3-dione (R6) [19]. Yang et al. [10] investigated the decomposition of butane-2,3-dione in a shock tube at 1200–1800 K and calculated the pressure-dependent rate constant of R3 via ab initio/Master Equation/VRC-TST method. Their calculated rate constant is in good agreement with their measured results. In summary, previous pyrolysis experiments of butane-2,3-dione were mainly conducted at low pressures (0.0008–0.6 bar) based on the review by Yang et al. [10] and focused on the sub-mechanism of butane-2,3-dione and the formation of primary products. Experimental data obtained at atmospheric and high pressures are quite scarce. Meanwhile, experimental studies on the formation of secondary products, mainly produced by carbon chain growth reactions related to CH_3 , are extremely limited.



In this work, the pyrolysis experiments of butane-2,3-dione were conducted in a flow reactor at 0.04 and 1 bar over the temperature range of 780–1520 K and in a jet-stirred reactor (JSR) at 10 bar between 650 and 1130 K, extending previous experimental conditions to atmospheric and high pressures. Several diagnostic methods, namely synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS), Fourier transform infrared spectroscopy (FTIR) and gas chromatography (GC), were introduced for the identification and quantification of both primary and secondary products. A pyrolysis model was developed to investigate the kinetics of butane-2,3-dione decomposition and the carbon chain growth from CH_3 to benzene (A1). It was validated against the present pyrolysis data, as well as those of C2–C4 fuels in the literature (see Table S1). Insight into the decomposition of butane-2,3-dione and the carbon chain growth from CH_3 was provided in the following discussions with a special emphasis on the pressure-dependencies.

2. Experimental method

2.1. Flow reactor pyrolysis

The flow reactor pyrolysis experiment was conducted at National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. The detailed descriptions on the VUV beamline (BL03U) dedicated to combustion research are available in our previous work [24]. The flow reactor is composed of an $\alpha\text{-Al}_2\text{O}_3$ ceramic tube (inner diameter 7.0 mm) and a home-made furnace. The catalytic effect of $\alpha\text{-Al}_2\text{O}_3$ is negligible according to the study of Narayanan et al. [25] and our previous work [26]. The length of the heating zone is 224 mm. A tungsten-rhenium (W-Re) thermocouple is positioned at the middle region of the outer wall to monitor and control the reactor temperature, while the temperature distributions along the centerline of the reactor were measured by an S-type thermocouple [27,28]. The complete temperature profiles are provided in the *Supplementary Materials*. Each temperature profile is named by its maximum temperature. Butane-2,3-dione (purity 98%, provided by Macklin) was injected into a vaporizer through a liquid flow controller. The total flow rate of the butane-2,3-dione/Ar mixture (2.31%/97.69%) is 1300 standard cubic centimeters per minute (SCCM) at both 0.04 and 1 bar. The pyrolysis products were sampled at 10 mm downstream from the outlet of the flow reactor by a quartz nozzle and measured with SVUV-PIMS method. The mass resolving power ($m/\Delta m$) of the mass spectrometer is more than 2000 [24]. Previously reported methods were used to identify and quantify the pyrolysis products [28]. The photoionization cross sections (PICSs) adopted in the quantification process are referred to our online database [29]. Generally, the uncertainties are $\pm 10\%$ for butane-2,3-dione and CO, $\pm 25\%$ for stable species with PICS available in the literature, and a factor of 1.5 for CH_2CO with a larger PICS uncertainty [30]. The mole fractions of the measured species are provided in the *Supplementary Materials*.

2.2. Jet-stirred reactor pyrolysis

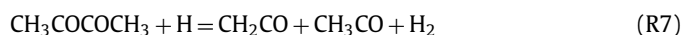
The JSR pyrolysis experiment was carried out at C.N.R.S.-I.N.S.I.S in Orléans, France. The detailed descriptions on the experimental apparatus can be found in our previous work [31]. Briefly, the reactor consists of a fused silica sphere with a volume of $\sim 38 \text{ cm}^3$. It is settled inside a stainless-steel jacket to resist high pressure. Liquid butane-2,3-dione (purity 97%, provided by Sigma-Aldrich) and auxiliary N_2 with high purity were introduced into the reactor through a heated vaporizer and then mixed with the primary N_2 and O_2 . The pressure inside the reactor was kept at 10 bar. The initial mole fraction of butane-2,3-dione was 2000 ppm and the flow rates of fuel and N_2 were adjusted at each temperature to keep a constant residence time (i.e. 700 ms in this work). Stirring was achieved by four nozzles with 1 mm orifice inside the sphere. The reacting mixtures were sampled using a low-pressure sonic probe. The detection of CO was performed with online FTIR, while other hydrocarbon and oxygenated products were quantified with GC. Two GCs equipped with flame ionization detectors (FID) were used in this work. One is equipped with a DB624 column to quantify oxygenated compounds (CH_2CO , CH_3CHO , CH_3COCH_3 and $\text{CH}_3\text{COCOCH}_3$), while the other one is equipped with a CP- $\text{Al}_2\text{O}_3/\text{KCl}$ column to quantify hydrocarbons (CH_4 , C_2H_2 , C_2H_4 , C_2H_6 , propyne, C_3H_6 , C_3H_8 and A1). Identification of the products was achieved by GC combined with mass spectrometry (MS). Besides, H_2 was quantified by a GC equipped with a CP-CarboPLOT P7 column and a thermal conductivity detector (TCD). The uncertainties are within $\pm 15\%$ for most of the JSR pyrolysis products, except for CH_2CO which is estimated to be a factor of 2. The mole fractions of the measured species can be found in the *Supplemental Materials*.

3. Kinetic modeling

The present model was developed based on our recent models of methanol [32], CH₃CHO [33] and butane isomers [34]. The sub-mechanisms of butane-2,3-dione and CH₂CO were mainly referred to the recent work from Christensen and Konnov [35,36]. The sub-mechanisms of ethane (C₂H₆), ethylene (C₂H₄), acetylene (C₂H₂), propane (C₃H₈), propene (C₃H₆), propyne (PC₃H₄), allene (AC₃H₄), 1,3-butadiene (C₄H₆) and vinylacetylene (C₄H₄), as well as the sub-mechanism of benzene formation, were carefully examined and developed in this work based on the evaluation of previous experimental and theoretical calculation studies on elementary reactions. Since the reactions related to the carbon chain growth are pressure-dependent, recent theoretical calculation results for them are preferentially considered, especially those validated against experimental measurements, in order to better characterize the pressure-dependent effects and the competition between the thermally-activated and chemically-activated reactions. Detailed descriptions of the evaluation of kinetic parameters can be found in the *Supplementary Materials*. Note that the order of reactions in the main text is shown in Table 1 which lists the key reactions discussed in this work. The thermodynamic data were referred to recent models for butane-2,3-dione [35], ketene [36] and butane isomers [34]. The files for the kinetic reaction mechanism and thermodynamic data are provided in the *Supplementary Materials*.

3.1. Sub-mechanisms of butane-2,3-dione and ketene

Yang et al. [10] investigated the unimolecular decomposition reaction of butane-2,3-dione in a shock tube at 1200–1800 K and various pressures (55, 120 and 225 Torr). They found the measured rate constant for R3 is pressure-dependent, which agrees well with their theoretical calculation results. For the modeling purpose, their calculated rate constants were fitted to the Troe format over 1200–1800 K and 55–220 Torr. Note that the high pressure limit (HPL) rate constant reported in the main text of their paper is wrong, while that reported in the abstract is right according to Fig. 3 of their paper, where the HPL rate constant was plotted. In a recent modeling work for butane-2,3-dione, Christensen and Konnov [35] adopted the correct HPL rate constant in the abstract of Yang et al. [10]. However, they used an incorrect low pressure limit (LPL) rate constant. This was checked in this work by plotting these different rate expressions versus the measurements of Yang et al. [10] at 55, 120 and 225 Torr. It is found that the reported LPL rate constant of R3 in the main text of Yang et al. [10] can agree with their measured results [10], while that modified by Christensen and Konnov [35] cannot. Therefore, in the present model, the HPL and LPL rate constants were adopted from the results reported in the abstract and the main text of Yang et al. [10], respectively. Yang et al. [10] also optimized the rate constant of R5 based on their experimental data. Their optimized value was incorporated in the present model. For the rate constant of R7, the estimated result by Yang et al. [10] was adopted, which was based on the analogy to CH₃COCH₃ + H = CH₃COCH₂ + H₂ with the consideration of differences in bond dissociation energy between butane-2,3-dione and CH₃COCH₃. Other reaction pathways involved in butane-2,3-dione pyrolysis are not as important as R3, R5 and R7. Their rate constants were taken from the recent modeling work of Christensen and Konnov [35].



Besides CH₃, CH₂CO is another important intermediate in butane-2,3-dione pyrolysis. In the present model, the sub-mechanism of CH₂CO was mainly referred to the recent modeling

work of Christensen and Konnov [35,36], in which the sub-mechanism of CH₂CO was updated and validated against various experimental data. Under the present experimental conditions, R8 is particularly important for the consumption of CH₂CO. However, the rate constant of R8 has large uncertainties [39,56–58]. In the present model, theoretical calculation results from a very recent work [39] were adopted for R8 and other competitive reaction channels on the potential of CH₂CO + CH₃. R9 is another important reaction and its rate constant was taken from a previous theoretical calculation work [38].



3.2. Simulation method

Based on the study of Rasmussen et al. [59], our flow reactor under the present investigated conditions satisfies the criteria of the Peclet number and the vessel dispersion. Therefore, simulations of the flow reactor experiments were performed using the Plug Flow Reactor module in the Chemkin Pro software [60] with the measured centerline temperature profiles as input parameters. For the JSR simulation, the Perfectly Stirred Reactor module in the Chemkin Pro software [60] was used with the problem type of “Fixing gas temperature”. The residence time was used as an input parameter for the JSR simulation instead of setting volume and inlet flow rate.

4. Results and discussion

4.1. Decomposition of butane-2,3-dione

The measured and predicted mole fractions of butane-2,3-dione, CO, CH₄ and CH₂CO in both flow reactor and JSR pyrolysis are presented in Fig. 1. As shown in Fig. 1(a) and (e), the present model could capture the temperature windows of fuel conversion at 0.04 and 1 bar within the experimental uncertainties, while it predicts a slightly faster than the measured conversion rate of butane-2,3-dione at 10 bar. Thus the production rates of CO, CH₄ and CH₂CO in JSR pyrolysis are accordingly overestimated, as shown in Fig. 1(e) and (f).

According to the rate of production (ROP) analysis and sensitivity analysis, the unimolecular decomposition reaction of butane-2,3-dione (R3), H abstraction reactions of butane-2,3-dione by CH₃ and H (i.e. R5 and R7) are the main consumption channels of butane-2,3-dione. The total contribution of R3, R5 and R7 to the consumption of butane-2,3-dione can reach more than 90%, as shown in Fig. 2(a)–(c). Among them, R3 and R5 are more important than R7, since R7 only has a minor contribution at 10 bar and becomes negligible at low and atmospheric pressures. The contribution of R3 increases with the increasing fuel conversion under the three investigated pressures while that of R5 presents an opposite trend. As we know, the fuel conversion increases with the increasing temperature. Compared with R5, R3 is a barrierless reaction and breaking C–C bond needs more energy to occur. Higher temperature provides more energy that makes R3 easier to occur. Besides, the relative contribution of R3 and R5 are different under three pressures. At low pressure, R3 is always the predominant consumption pathway of the fuel, while the contribution of R5 increases with increasing pressure and becomes competitive to that of R3 under atmospheric and high pressures. As shown in Fig. 2(d)–(f), the sensitivity analysis of butane-2,3-dione was conducted under fuel conversions of 10%, 20% and 70% at three pressures. At 0.04 bar, R3 is always the predominant sensitive reaction.

Table 1Key reactions discussed in this work, units: s^{-1} , $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$, cal/mol.

No.	Reactions	A	n	E	Reference
R1	$\text{CH}_3 + \text{CH}_3 (+\text{M}) = \text{C}_2\text{H}_6 (+\text{M})$	8.87×10^{16}	−1.16	774.5	[10]
	Low/	3.74×10^{50}	−9.93	7389/	
	Troe/	0.7548 158	32,828 46,564/		
R2	$\text{PC}_3\text{H}_4 + \text{H} = \text{C}_2\text{H}_2 + \text{CH}_3$	1.0	0.0	0.0	[37]
	Plog/0.0113	2.44×10^{10}	1.04	3980/	
	Plog/0.0395	3.89×10^{10}	0.99	4114/	
	Plog/1	3.46×10^{12}	0.44	5463/	
	Plog/10	1.72×10^{14}	−0.01	7134/	
	Plog/100	-7.40×10^{30}	−5.54	12108/	
	Plog/100	1.90×10^{15}	−0.29	8306/	
R3	$\text{CH}_3\text{COCOCH}_3 (+\text{M}) = 2\text{CH}_3\text{CO} (+\text{M})$	1.30×10^{28}	−3.4	74840	[10]
	Low/	1.63×10^{48}	−9.17	51830/	
	Troe/	0.52 3074	496 4100/		
R4	$\text{CH}_3\text{CO} = \text{CH}_3 + \text{CO}$	1.0	0.0	0.0	[38]
	Plog/0.01	6.88×10^{14}	−1.97	14585/	
	Plog/0.025	2.40×10^{15}	−2.00	14805/	
	Plog/0.1	1.96×10^{16}	−2.09	15197/	
	Plog/1	6.45×10^{18}	−2.52	16436/	
	Plog/10	8.18×10^{19}	−2.55	17263/	
	Plog/100	1.26×10^{20}	−2.32	18012/	
R5		1.41	4.0	7900	[10]
	$\text{CH}_3\text{COCOCH}_3 + \text{CH}_3 = \text{CH}_2\text{CO} + \text{CH}_3\text{CO} + \text{CH}_4$				
R6	$\text{CH}_3\text{COCOCH}_3 + \text{CH}_3 = \text{CH}_3\text{COCH}_3 + \text{CH}_3\text{CO}$	2.51×10^{10}	0.0	6400	[18]
R7	$\text{CH}_3\text{COCOCH}_3 + \text{H} = \text{CH}_2\text{CO} + \text{CH}_3\text{CO} + \text{H}_2$	9.18×10^7	2.0	5000	[10]
R8	$\text{CH}_2\text{CO} + \text{CH}_3 = \text{C}_2\text{H}_5 + \text{CO}$	1.24×10^5	2.29	10642	[39]
R9	$\text{CH}_2\text{CO} + \text{H} = \text{CH}_3 + \text{CO}$	7.80×10^8	1.45	2780	[38]
R10	$\text{C}_2\text{H}_6 + \text{CH}_3 = \text{C}_2\text{H}_5 + \text{CH}_4$	35	3.44	10384	[7]
R11	$\text{C}_2\text{H}_6 + \text{H} = \text{C}_2\text{H}_5 + \text{H}_2$	7.35×10^3	3.1	5340	[40]
	Duplicate				
	$\text{C}_2\text{H}_6 + \text{H} = \text{C}_2\text{H}_5 + \text{H}_2$	3.26×10^{14}	0.0	13666.8	
	Duplicate				
R12	$\text{C}_2\text{H}_4 + \text{H} (+\text{M}) = \text{C}_2\text{H}_5 (+\text{M})$	1.367×10^9	1.463	1355	[41]
	Low/	2.03×10^{39}	−6.64	5769/	
	Troe/	−0.569 299	−9147 152.4/		
	$\text{N}_2/1.2/\text{CO}/1.5/\text{H}_2/2/\text{CO}_2/3/\text{H}_2\text{O}/10/$				
R13	$\text{C}_2\text{H}_4 + \text{H} = \text{C}_2\text{H}_3 + \text{H}_2$	5.07×10^7	1.93	12950	[42]
R14	$\text{C}_2\text{H}_4 + \text{CH}_3 = \text{C}_2\text{H}_3 + \text{CH}_4$	9.75×10^2	2.947	15148	[43]
	Duplicate				
	$\text{C}_2\text{H}_4 + \text{CH}_3 = \text{C}_2\text{H}_3 + \text{CH}_4$	8.13×10^{-5}	4.417	8835.8	
	Duplicate				
R15	$\text{C}_2\text{H}_2 + \text{H} (+\text{M}) = \text{C}_2\text{H}_3 (+\text{M})$	1.0	0.0	0.0	[44]
	TCHEB/	500	2000/		
	PCHEB/	0.001	100/		
	CHEB/	7 4	10.6	2.25	−0.06/
	CHEB/	−0.335	0.248	0.142	−0.35/
	CHEB/	-5.97×10^{-3}	3.36×10^{-3}	8.7×10^{-3}	$-9.5 \times 10^{-3}/$
	CHEB/	-5.48×10^{-3}	-1.0×10^{-3}	-2×10^{-4}	$-2.2 \times 10^{-3}/$
	CHEB/	-1.28×10^{-3}	-2.2×10^{-3}	-3×10^{-4}	$-3.7 \times 10^{-4}/$
	CHEB/	8.88×10^{-4}	2.4×10^{-4}	1.3×10^{-4}	$1.5 \times 10^{-5}/$
R16	$\text{C}_3\text{H}_8 (+\text{M}) = \text{CH}_3 + \text{C}_2\text{H}_5 (+\text{M})$	1.29×10^{37}	−5.84	97380	[12]
	Low/	5.64×10^{74}	−15.74	98714/	
	Troe/	0.31 50	3000 9000/		
R17	$\text{H}_2/2.0/\text{H}_2\text{O}/6.0/\text{CH}_4/2.0/\text{CO}/1.5/\text{CO}_2/2.0/\text{C}_2\text{H}_6/3.0/\text{HE}/0.7/$				
	$\text{C}_2\text{H}_4 + \text{CH}_3 = \text{NC}_3\text{H}_7$	1.0	0	0	[43]
	Plog/0.001	8.66×10^{48}	−12.51	18206/	
	Plog/0.04	1.06×10^{49}	−12.03	20001/	
	Plog/1	7.67×10^{47}	−11.17	22366/	
	Plog/10	1.81×10^{45}	−10.03	23769/	
	Plog/100	2.04×10^{40}	−8.25	24214/	
	Duplicate				
	Plog/0.001	1.12×10^{43}	−11.30	13080/	
	Plog/0.04	7.28×10^{39}	−9.88	13164/	
	Plog/1	2.60×10^{33}	−7.46	12416/	
	Plog/10	3.85×10^{27}	−5.38	11455/	
	Plog/100	1.66×10^{21}	−3.17	10241/	
	Duplicate				
R18	$\text{C}_3\text{H}_6 + \text{H} = \text{C}_2\text{H}_4 + \text{CH}_3$	1.0	0	0	[43]
	Plog/0.0013	1.54×10^9	1.35	2542/	
	Plog/0.10	7.88×10^{10}	0.87	3599/	
	Plog/1	2.67×10^{12}	0.47	5431/	
	Plog/10	9.25×10^{22}	−2.60	12898/	
	Plog/100	1.32×10^{23}	−2.42	16500/	
	Duplicate				
	Plog/10	1.24×10^5	2.52	3679.1/	
	Plog/100	2.51×10^3	2.91	3980.9/	

(continued on next page)

Table 1 (continued)

No.	Reactions	A	n	E	Reference
R19	Duplicate				
	$C_3H_6 (+M) = CH_3 + C_2H_3 (+M)$	1.0	0.0	0.0	[45]
	TCHEB/ 300	2500/			
	PCHEB/ 0.001	100/			
	CHEB/ 7 4	−22.8	0.62	−0.11	0.014/
	CHEB/ 29.5	0.965	−0.107	-6×10^{-3}	−0.82/
	CHEB/ 0.52	0.0186	−0.024	−0.47	0.188/
	CHEB/ 0.059	−0.013	−0.24	0.0262	0.0407/
	CHEB/ 1.66×10^{-3}	−0.106	−0.023	0.0133	6.29×10^{-3} /
R20	$AC_3H_5 + H(+M) = CH_3 + C_2H_3 (+M)$	1.0	0.0	0.0	[45]
	TCHEB/ 300	2500/			
	PCHEB/ 0.001	100/			
	CHEB/ 7 4	10.64	−1.61	−0.11	0.013/
	CHEB/ 3.81	1.268	−0.054	−0.021	0.106/
	CHEB/ 0.478	0.1	−0.024	−0.162	0.019/
	CHEB/ 0.089	8.24×10^{-3}	−0.106	−0.091	0.0179/
	CHEB/ 0.0185	−0.0335	−0.060	−0.018	8.43×10^{-3} /
	CHEB/ -3.7×10^{-3}	−0.0206	−0.019	-2.3×10^{-3} /	
R21	$C_3H_6(+M) = AC_3H_5 + H(+M)$	1.0	0.0	0.0	[45]
	TCHEB/ 300	2500/			
	PCHEB/ 0.001	100/			
	CHEB/ 7 4	−21.02	0.399	−0.056	3.25×10^{-3} /
	CHEB/ 27.5	0.6987	−0.083	1.1×10^{-3}	−0.663/
	CHEB/ 0.469	−0.0243	-6.84×10^{-3}	−0.428	0.2345/
	CHEB/ 0.0186	-7.85×10^{-3}	−0.2399	0.0773	0.0280/
	CHEB/ -2.6×10^{-3}	−0.1184	5.03×10^{-3}	0.0168	2.29×10^{-3} /
	CHEB/ −0.0556	−0.01668	2.06×10^{-3}	3.48×10^{-3} /	
R22	$C_4H_6 + H = C_2H_4 + C_2H_3$	1.0	0	0	[46]
	Plog/0.01	1.35×10^{73}	−15.58	80,080/	
	Plog/0.1	6.57×10^{40}	−7.68	25,930/	
	Plog/1	1.88×10^{40}	−7.28	29,420/	
	Plog/10	9.18×10^{35}	−5.82	31,360/	
	Plog/100.	1.34×10^{34}	−5.15	35,320/	
	Duplicate				
	Plog/0.01	1.20×10^{25}	−3.06	16,390/	
	Plog/0.1	1.04×10^{14}	0.24	13,800/	
	Plog/1	1.19×10^{10}	1.38	13,630/	
	Plog/10	1.16×10^8	1.82	13,720/	
	Plog/100	2.83×10^9	4.12	12,500/	
	Duplicate				
	$C_4H_7-1 = C_2H_5 + C_2H_2$	1.0	0	0	[46]
	Plog/0.01	1.02×10^{75}	−19.59	53,540/	
R23	Plog/0.1	9.89×10^{70}	−17.94	54,010/	
	Plog/1	5.33×10^{62}	−15.02	52,920/	
	Plog/10	2.91×10^{55}	−12.49	52,140/	
	Plog/100.	1.05×10^{49}	−10.29	51,620	
	Duplicate				
	Plog/0.01	1.13×10^{43}	−9.94	39,680/	
	Plog/0.1	1.17×10^{38}	−8.14	38,920/	
	Plog/1	2.29×10^{34}	−6.84	38,250/	
	Plog/10	2.64×10^{31}	−5.81	37,730/	
	Plog/100	9.18×10^{25}	−3.93	36,620/	
	Duplicate				
	$C_4H_6-1 + H = C_2H_5 + C_2H_2$	1.0	0	0	[46]
	Plog/0.01	2.42×10^{16}	−0.48	25,560/	
	Plog/0.1	3.31×10^{37}	−7.29	29,090/	
R24	Plog/1	4.67×10^{31}	−5.14	29,630/	
	Plog/10	2.10×10^{35}	−6.10	34,650/	
	Plog/100.	3.15×10^{32}	−4.96	37,190/	
	Duplicate				
	Plog/0.01	2.54×10^{19}	−1.66	21,020/	
	Plog/0.1	1.02×10^{14}	0.13	20,710/	
	Plog/1	1.22×10^{10}	1.28	20,510/	
	Plog/10	4.44×10^8	1.72	21,310/	
	Plog/100	1.90×10^6	2.29	21,380/	
	Duplicate				
	$C_4H_6 = C_3H_3 + CH_3$	1.0	0	0	[47]
	Plog/0.01	1.63×10^{80}	−19.09	123,600/	
	Plog/0.1	9.10×10^{76}	−17.82	126,500/	
	Plog/1	2.63×10^{70}	−15.65	127,800/	
	Plog/10	1.72×10^{60}	−12.51	127,200/	
	Plog/100	1.36×10^{45}	−8.06	123,200/	
R25	$C_2H_2 + C_2H_3 = C_4H_4 + H$	1.0	0	0	[48]
	Plog/0.01	1.04×10^{26}	−3.67	15,007/	
	Plog/0.1	7.30×10^{28}	−4.41	17,570/	

(continued on next page)

Table 1 (continued)

No.	Reactions	A	n	E	Reference
R27	Duplicate	Plog/1	1.72×10^{27}	-3.72	19,301/
		Plog/10	6.31×10^{25}	-3.16	21,319/
		Plog/100.	1.06×10^{39}	-6.85	33,640/
	Duplicate	Plog/0.01	2.77×10^{12}	0.57	11,814/
		Plog/0.1	5.73×10^{14}	-0.0242	15,274/
		Plog/1	3.08×10^{26}	-3.0312	34,136/
	Duplicate	Plog/10	1.01×10^{68}	-13.764	95,268/
		Plog/100	7.31×10^{12}	0.546	16,743/
	Duplicate	$H_2CC + C_2H_2 = IC_4H_3 + H$	1.0	0	0
		Plog/0.01	2.57×10^{119}	-27.56	176,200/
		Plog/1	9.84×10^{27}	-3.99	35,280/
	Duplicate	Plog/10	5.48×10^{30}	-4.63	43,210/
		Plog/0.01	1.03×10^{22}	-2.51	24,720/
		Plog/1	1.60×10^{71}	-18.29	44,490/
		Plog/10	1.08×10^{18}	-1.53	28,780/
R28	Duplicate	$C_4H_{10} (+M) = C_2H_5 + C_2H_5 (+M)$	2.72×10^{15}	0.0	75,605
		Low/	4.72×10^{18}	0.0	49,575/
		Troe/	0.28 0.1	1500 50,000/	
R29	Duplicate	$PC_4H_9 = C_2H_4 + C_2H_5$	1.0	0	0
		Plog/0.01	1.11×10^{39}	-9.05	31,891/
		Plog/1	7.47×10^{43}	-9.67	38,722/
R30	Duplicate	Plog/10	2.06×10^{39}	-7.97	38,955/
		SC ₄ H ₉ = C ₂ H ₄ + C ₂ H ₅	1.0	0	0
		Plog/0.01	4.12×10^{27}	-5.94	39,859/
R31	Duplicate	Plog/1	4.54×10^{42}	-9.54	50,839/
		Plog/10	1.06×10^{49}	-10.9	58,899/
		$C_4H_8-1 + H = C_2H_4 + C_2H_5$	1.0	0	0
R32	Duplicate	Plog/0.01	5.56×10^6	1.83	5802/
		Plog/1	9.47×10^{16}	-1.03	13,413/
		Plog/10	4.50×10^{28}	-4.24	23,618/
R33	Duplicate	Plog/0.01	8.06×10^7	1.71	2522/
		Plog/1	6.02×10^{15}	-0.49	8452/
		Plog/10	7.58×10^{21}	-2.14	14,545/
R34	Duplicate	$C_4H_8-2 + H = C_2H_4 + C_2H_5$	1.0	0	0
		Plog/0.01	1.92×10^7	1.77	6443/
		Plog/1	3.01×10^{17}	-1.09	14,023/
R35	Duplicate	Plog/10	1.88×10^{29}	-4.33	24,297/
		$C_3H_3 + C_3H_3 = A1$	1.0	0	0
		Plog/0.04	1.29×10^{69}	-16.73	27,864/
R36	Duplicate	Plog/1	3.16×10^{55}	-12.55	22,264/
		Plog/10	3.89×10^{50}	-11.01	20,320/
		Plog/0.04	7.07×10^{39}	-8.82	7049/
R37	Duplicate	$C_3H_3 + C_3H_3 = \text{Fulvene}$	1.0	0	0
		Plog/0.04	2.34×10^{69}	-17.02	25,864/
		Plog/1	5.49×10^{62}	-14.73	25,602/
R38	Duplicate	Plog/10	5.62×10^{60}	-13.93	27,093/
		Plog/0.04	1.74×10^{44}	-10.31	7992/
		Plog/1	3.01×10^{35}	-7.37	5963/
R39	Duplicate	Plog/10	7.97×10^{29}	-5.5	4665/
		$C_3H_3 + AC_3H_5 = \text{fulvene} + 2H$	1.0	0	0
		Plog/0.04	2.29×10^{43}	-9.123	20,504/
R40	Duplicate	Plog/1	1.88×10^{38}	-7.526	23,878/
		Plog/10	4.44×10^{23}	-3.364	18,249/
		$NC_4H_5 + C_2H_2 = A1 + H$	1.0	0	0
R41	Duplicate	Plog/0.01	1.37×10^{16}	-1.00	8896/
		Plog/1	1.385×10^{16}	-1.00	8900/
		Plog/10	1.69×10^{16}	-1.03	8967/
R42	Duplicate	$NC_4H_5 + C_2H_2 = \text{fulvene} + H$	1.0	0	0
		Plog/0.01	1.52×10^{15}	-0.76	8767/
		Plog/1	4.63×10^{15}	-0.89	9142/
R43	Duplicate	Plog/10	1.74×10^{19}	-1.86	12,383/
		$IC_4H_5 + C_2H_2 = \text{fulvene} + H$	1.0	0	0
		Plog/0.01	6.50×10^{24}	-3.44	20,319/
R44	Duplicate	Plog/1	6.80×10^{24}	-3.45	20,337/
		Plog/10	9.70×10^{25}	-3.76	21,326/

(continued on next page)

Table 1 (continued)

No.	Reactions	A	n	E	Reference
R39	Fulvene + H = A1 + H	1.0	0.0	0.0	[55]
	Plog/0.001	4.94×10^{18}	-1.28	5411/	
	Plog/0.01	2.15×10^{22}	-2.28	8429/	
	Plog/0.1	5.60×10^{26}	-3.47	12,818/	
	Plog/1	1.66×10^{25}	-2.99	13,691/	
	Plog/1.312	5.06×10^{25}	-3.12	14,226/	
	Plog/13.12	2.20×10^{27}	-3.48	19,199/	

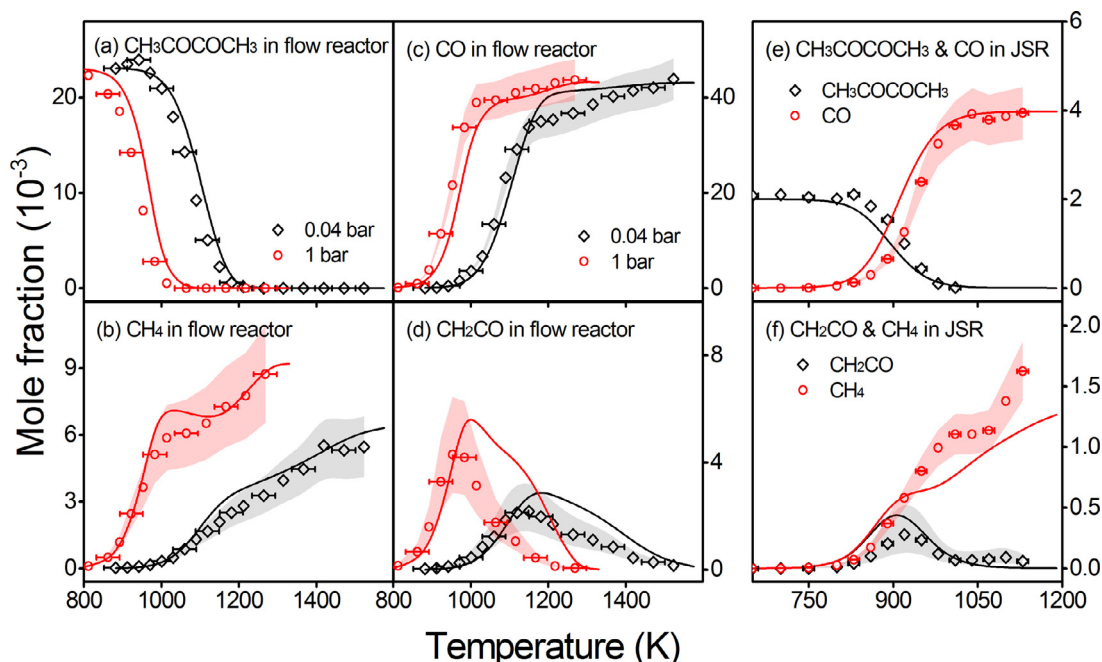


Fig. 1. Measured (symbols) and predicted (lines) mole fraction profiles of (a) butane-2,3-dione, (b) CH₄, (c) CO and (d) CH₂CO in the flow reactor pyrolysis at 0.04 and 1 bar; (e) butane-2,3-dione and CO and (f) CH₂CO and CH₄ in the JSR pyrolysis at 10 bar. The mole fraction and temperature uncertainties are shown with shadows and bars, respectively.

However, at 1 and 10 bar, although R3 is still a very sensitive reaction, especially at high fuel conversion, the sensitivity of R5 cannot be neglected, especially under low fuel conversion and high pressure conditions.

Based on the above analysis, R3 plays a critical role in accurately predicting the fuel consumption from low to high pressures. In the present model, the pressure-dependent rate constant of R3 was adopted from the work of Yang et al. [10], where the calculated results were constrained by their measurements between 0.072 and 0.296 bar and fitted to the Troe format over the temperature range of 1200–1800 K and the pressure range of 0.072–0.296 bar. Thus, the rate constant of R3 from Yang et al. [10] can predict the present experimental results at 0.04 and 1 bar, while at 10 bar which is far beyond the investigated pressures by Yang et al. [10], it seems to be fast and leads to the slight over-prediction of the butane-2,3-dione conversion. Further work on the pressure-dependence of the rate constant of R3 at wider temperature and pressure ranges will be desired.

4.2. Formation of oxygenated species

CH₂CO is considered to be the most important oxygenated species in the pyrolysis of butane-2,3-dione and can be formed through R5 and R7 [16–23]. In this work, CH₂CO was measured in both flow reactor and JSR pyrolysis. As can be seen from Fig. 1(d), the maximum mole fraction of CH₂CO at 1 bar is apparently higher than that at 0.04 bar. This is because H abstraction reactions of

butane-2,3-dione (i.e. R5 and R7) at 1 bar are more important than at 0.04 bar. As shown in Fig. 1(d) (solid lines), the production of CH₂CO at both 0.04 and 1 bar can be captured by the present model, while its consumption at 1 bar is under-estimated. According to the ROP analysis, the most important consumption pathway of CH₂CO is R8 under the present CH₃ abundant conditions. This reaction is competitive with R1 and converts CH₃ to C₂H₅, playing an important role in accurately predicting the radical pool. Since the formation of C₂H₆, C₂H₄ and C₃H₈ is directly related to the local concentration of CH₃ and/or C₂H₅, their mole fractions are also sensitive to R8. However, the rate constant of R8 has large uncertainties. At 1000 K, the newly calculated value by Semenikhin et al. [39] is almost 10 times smaller than that proposed by Borisov et al. [58], which was adopted in a recent modeling work for CH₂CO [36]. The present model with the rate constant from Semenikhin et al. [39] can better capture the formation of C2 and C3 hydrocarbons at 10 bar while over-predict the formation of CH₂CO at 1 bar. Considering the rate source from Semenikhin et al. [39] is based on recent theoretical calculations at high level and the experimental uncertainties of CH₂CO are great, the rate constant from Semenikhin et al. [39] for R8 was adopted in this model. Besides, the present model was also validated against the pyrolysis data of CH₂CO found in the literature [57,61–63], as shown in Figs. S1–S4 in the Supplementary Materials. Compared with previous models, i.e. Konnov model [35], AramcoMech 2.0 model [51] and HP model [64], the present model could better predict the decomposition of CH₂CO and the formation of products.

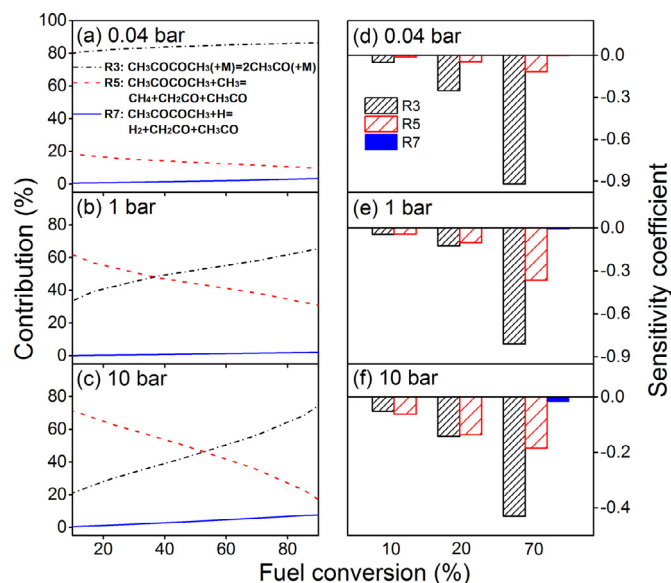


Fig. 2. Left: ROP analysis of butane-2,3-dione versus the fuel conversion at (a) 0.04 bar, (b) 1 bar and (c) 10 bar; Right: Sensitivity analysis of butane-2,3-dione under fuel conversion ratios of 10%, 20% and 70% at (d) 0.04 bar, (e) 1 bar and (f) 10 bar.

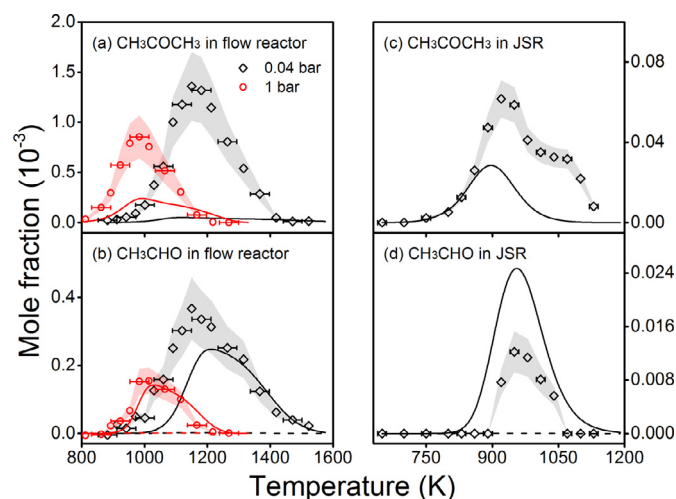
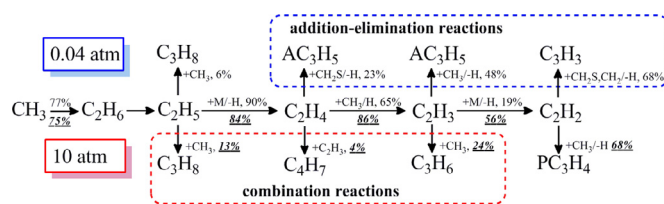


Fig. 3. Measured (symbols) and predicted (lines) mole fraction profiles of (a) CH_3COCH_3 , (b) CH_3CHO in the flow reactor pyrolysis of butane-2,3-dione at 0.04 and 1 bar and (c) CH_3COCH_3 and (d) CH_3CHO in the JSR pyrolysis of butane-2,3-dione at 10 bar. Solid and dashed lines represent the present model with and without R42, respectively. The mole fraction and temperature uncertainties are shown with shadows and bars, respectively.

CH_3COCH_3 is another important oxygenated product in the pyrolysis of butane-2,3-dione [20]. It can be formed through the addition-elimination reaction of butane-2,3-dione + CH_3 (R6) according to Hole and Mulcahy [19]. Their measured rate constant at 677–776 K and 0.6–45 Torr is in good agreement with Knoll et al. [18] at 648–690 K and 43–183 Torr. However, the present model with the measured rate constant of R6 by Knoll et al. [18] greatly under-predicts CH_3COCH_3 at all investigated pressures, and the discrepancies become greater as the pressure decreases, as shown in Fig. 3(a) and (c). Recognizing that addition-elimination reactions compete with addition reactions, high temperature/low pressure conditions make addition-elimination reactions more favorable. Therefore, a pressure-dependent rate constant for R6, which is not available in the literature, will be more reliable than the one without pressure-dependence adopted in the



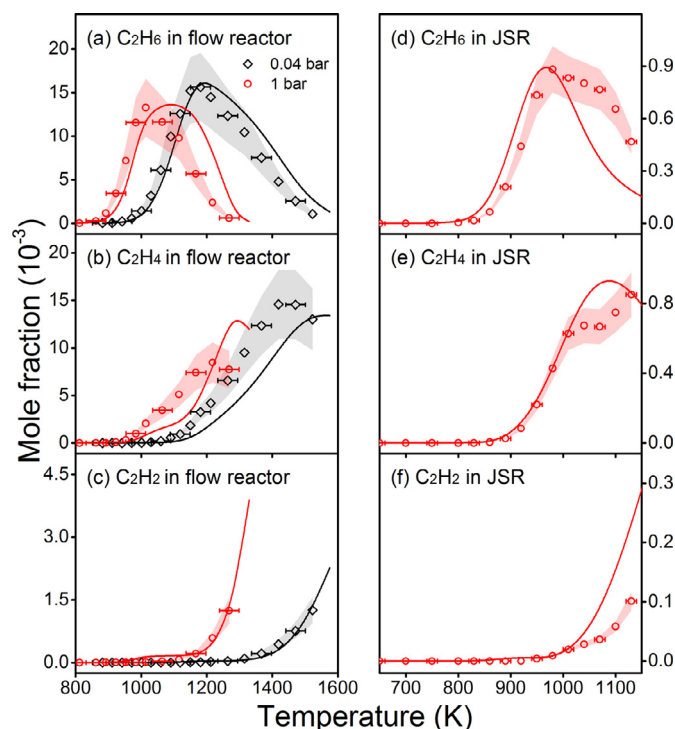


Fig. 5. Measured (symbols) and predicted (lines) mole fraction profiles of (a) C₂H₆, (b) C₂H₄ and (c) C₂H₂ in the flow reactor pyrolysis of butane-2,3-dione at 0.04 and 1 bar and (d) C₂H₆, (e) C₂H₄ and (f) C₂H₂ in the JSR pyrolysis of butane-2,3-dione at 10 bar. The mole fraction and temperature uncertainties are shown with shadows and bars, respectively.

pathways at 1 atm lay between those observed at 0.04 and 10 atm, which are not separately listed in this work.

4.3.1. C₂ pathways

The C₂ hydrocarbon products measured in this work include ethane, ethylene and acetylene. As shown in Fig. 5, the present model could reasonably predict the formation of C₂ hydrocarbon products measured in this work. According to the ROP analysis in Fig. 4, they are mainly formed via the following reaction sequence: $2\text{CH}_3 \rightarrow \text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5 \rightarrow \text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_3 \rightarrow \text{C}_2\text{H}_2$. The formation of ethane seems to be pressure-dependent according to our experimental and predicted results (see Fig. 5(a)). The maximum mole fraction of ethane at 0.04 bar is slightly higher than that at 1 bar, whereas an opposite trend is observed for CH₄ (see Fig. 1(b)). According to the ROP analysis, a possible reason could be that the contribution of H abstraction reactions by CH₃ at 0.04 bar is smaller than that at atmospheric and high pressures, making the relative contribution of its competitive reaction (i.e. R1) larger at 0.04 bar than those at 1 and 10 bar.

The self-combination reaction of CH₃ (R1) is not only crucial to accurately predict the formation of C₂ species in this work, but is also sensitive in C₂H₆ pyrolysis through the reverse direction. The predicted results of the present model and previous models [27,44,51] for the shock tube pyrolysis experiment of C₂H₆ at 1.664 bar and 1589 K [12] are presented in Fig. S5 in the Supplementary Materials. The present model can reasonably predict the formation of CH₃ in the pyrolysis of ethane, as well as AramcoMech 2.0 [51] and 2012 Zhang model [27]. However, CSM model [44] predicts a much slower CH₃ formation compared with the measured results. Besides, the sub-mechanism of C₂H₆ in the present model was also validated against other experimental targets of C₂H₆ pyrolysis in the literature [65–68]. The predicted results are presented in Figs. S6–S9 in the Supplementary Materials,

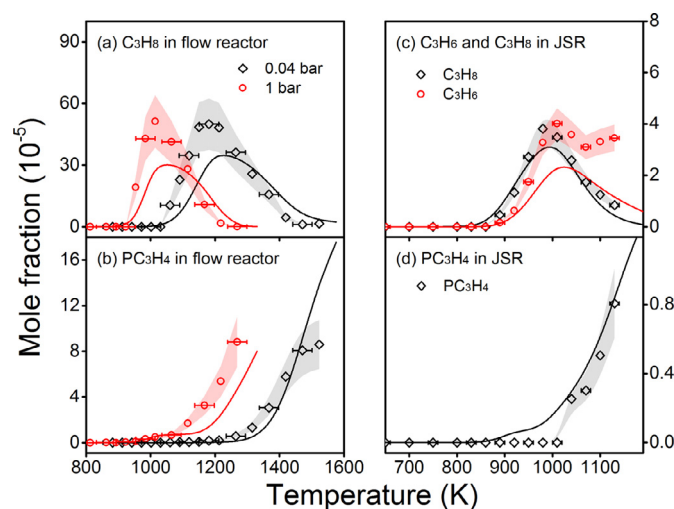


Fig. 6. The measured (symbols) and predicted (lines) mole fraction profiles of (a) C₃H₈, (b) PC₃H₄ in the flow reactor pyrolysis of butane-2,3-dione at 0.04 and 1 bar; (c) C₃H₈ and C₂H₆, (d) PC₃H₄ in the JSR pyrolysis of butane-2,3-dione at 10 bar. The mole fraction and temperature uncertainties are shown with shadows and bars, respectively.

covering the temperature range of 800–1900 K and the pressure range of 0.8–1000 bar. Generally, the present model can reasonably predict all the validation targets, greatly improving the predictions of our previous model [27].

4.3.2. C₃ pathways

As shown in Fig. 4, C₂H₅ mainly undergoes the β -C-H scission reaction (R12) to produce C₂H₄. Meanwhile, it can also combine with CH₃ to form C₃H₈ (R16), resulting in the growth of carbon chain. Figure 4 shows that R16 is more important at 10 bar than at 0.04 bar. High pressure condition promotes the occurrence of collision and therefore makes combination reactions more favorable at 10 bar than at 0.04 bar. The measured mole fraction profiles of C₃H₈ at 1 and 10 bar in this work provide additional data to validate the rate constant of R16 over 800–1100 K compared with previous validation targets listed in Table S1. The present model could capture the trends of C₃H₈ profiles measured in both this work (see solid lines in Fig. 6(a) and (c)) and previous shock tube pyrolysis work of C₃H₈ above 1100 K [12,69,70] (see Figs. S13–S14 in the Supplementary Materials). The deviation between the measured and predicted results of C₃H₈ in butane-2,3-dione pyrolysis could be explained by the fact that the rate constant of R8 has not been well studied and large uncertainties still exist. The subsequent H loss reaction sequence from C₃H₈, i.e. $\text{C}_3\text{H}_8 \rightarrow \text{IC}_3\text{H}_7 \rightarrow \text{C}_3\text{H}_6 \rightarrow \text{AC}_3\text{H}_5 \rightarrow \text{AC}_3\text{H}_4 \rightarrow \text{PC}_3\text{H}_4$, partially contributes to the formation of other C₃ hydrocarbon products measured in this work.

C₂H₃ is mainly produced through the H abstraction reactions of C₂H₄. The reactions between C₂H₃ and CH₃ can also lead to the formation of C₃ hydrocarbon products. At low pressure, the addition-elimination reaction between C₂H₃ and CH₃ (R20) is favored and allyl radical (AC₃H₅) and H can be generated. However, at high pressure, the combination reaction between C₂H₃ and CH₃ producing C₃H₆ (R19) is more important. R19 and R20 are thermally-activated and chemically-activated channels on the PES of C₃H₆, respectively. The chemically-activated pathway (R20) directly converts the initially formed (energized) C₃H₆ to bimolecular products (AC₃H₅ and H) before this adduct is collisionally stabilized to form C₃H₆. Under low pressure and high temperature conditions (e.g. at 0.04 bar and 1367 K), the collisional stabilization is slow, making R20 more favorable. In contrast, strong collision at

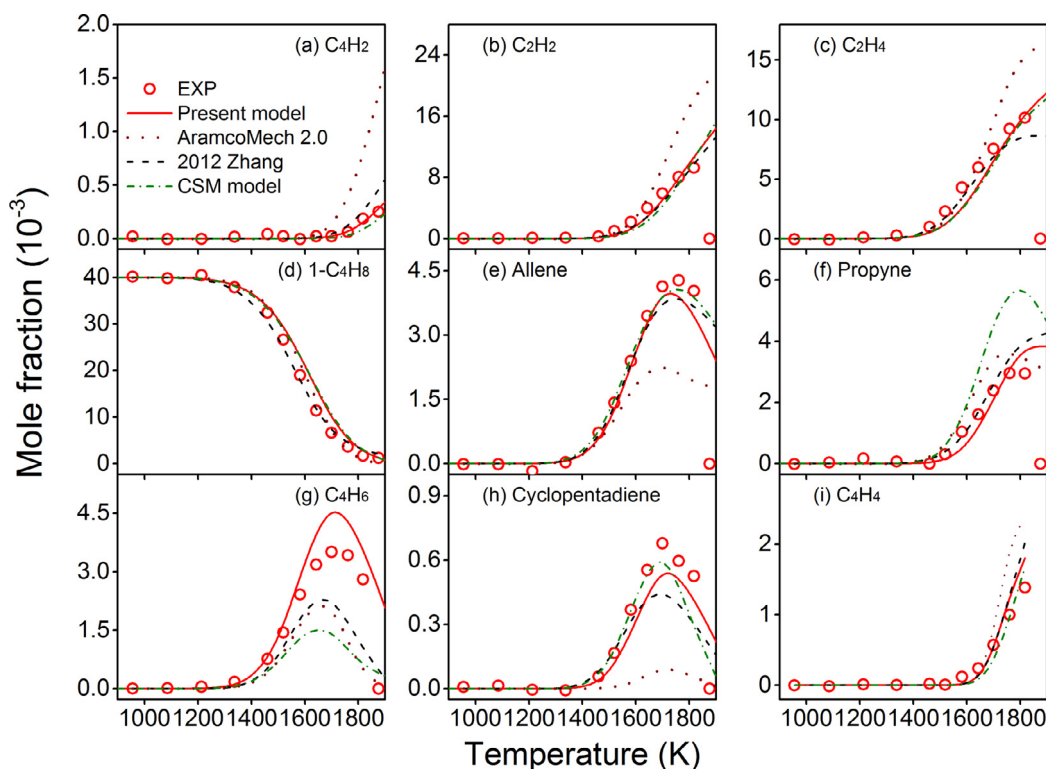


Fig. 7. Comparison between the measured [27] (symbols) and predicted (lines) results in the flow reactor pyrolysis of 1-butene (1-C₄H₈) at low pressures. The mixture is 4% 1-C₄H₈/96% Ar. Solid, dash dot, dotted, dashed lines represent the predicted results of present model, CSM model [44], AramcoMech 2.0 model [51] and 2012 Zhang model [27], respectively.

high pressure conditions (e.g. at 10 bar) favors the occurrence of R19. C₃H₆ was measured in the JSR pyrolysis of butane-2,3-dione at 10 bar with GC method, while could hardly be separated from CH₂CO in the flow reactor pyrolysis of butane-2,3-dione with SVUV-PIMS method due to its much lower concentration levels than CH₂CO. Therefore, only the measured result in the JSR pyrolysis experiment is presented. As shown in Fig. 6(c), the present model under-predicts the formation of C₃H₆. One possible reason is that the rate constant of R19 has received little attention and consequently has large uncertainties. The recent calculated result by Wang et al. [45] is 5–10 times larger than that calculated by Ye et al. [71] at 10 bar and 800–1200 K. However, previous pyrolysis experiments of C₃H₆ [72–74] could hardly provide enough constraint for R19 since R21 (C₃H₆ (+M) = AC₃H₅ + H (+M)) is the predominant channel in C₃H₆ pyrolysis. Fortunately, the measured mole fractions of C₃H₆ and C₂H₂ in this work are very sensitive to R19, because R19 is not only the major source of C₃H₆ but is also the most important competing reaction channel of R15, which contributes to the formation of C₂H₂. As shown in Fig. 6(c) and Fig. 5(f), the formation of C₃H₆ is under-estimated while that of C₂H₂ is over-predicted. Thus, a larger rate constant of R19 would be expected, which could improve the predictions of both C₃H₆ and C₂H₂. The present experimental data complement the validation data sets of C₃H₆, highlighting the significance of R19 which still needs further studies. In addition, the present model can reasonably predict previous experimental targets [72–74] of C₃H₆ pyrolysis, as shown in Figs. S15–S18 in the *Supplementary Materials*.

The addition-elimination reaction between C₂H₂ and CH₃ (R2) leads to the formation of PC₃H₄ and H. The measured and predicted mole fraction profiles of PC₃H₄ are presented in Fig. 6(b) and (d). The present model can reasonably predict the production of PC₃H₄ at 0.04 and 10 bar while under-estimates its formation at 1 bar. In order to better validate the sub-mechanism of PC₃H₄, the experimental targets involving AC₃H₄ and PC₃H₄ in the

literature [75] were selected and the present model can well predict them, as shown in Figs. S19 and S20 in the *Supplementary Materials*, which presents better performance than our previous model [27] and recent models [44,51].

4.3.3. C₄ pathways

The growth of carbon chain can also proceed through C1 + C3 or C2 + C2 pathways to form C₄ hydrocarbon products, i.e. CH₃ + AC₃H₅ → 1-C₄H₈ → methylallyl (SAXC₄H₇) → C₄H₆ and C₂H₄ + C₂H₃ → C₄H₆ + H (R22), respectively. However, these pathways are not as efficient as C1 + C2 pathways since C₄ species were only detected at trace levels. The C1 + C3 pathway plays an important role in the pyrolysis of 1-butene. The present model, 2012 Zhang [27] model and CSM model [44] can well predict the consumption of 1-butene and the formation of C₄ species in the flow reactor pyrolysis of 1-butene at low pressures [27], showing better performance than the AramcoMech 2.0 model [51], as shown in Fig. 7. R22 is important in the pyrolysis of C₂H₄ [76] and 1,3-butadiene (C₄H₆) [77–79]. The present model can not only reproduce the decomposition of C₂H₄ and the formation of C₄H₆ in C₂H₄ pyrolysis [76] (see Fig. 8), but also reasonably predict the consumption of fuel and the formation of decomposition products in C₄H₆ pyrolysis [77–79] (see Figs. S27–S34 in the *Supplementary Materials*).

Besides, the previous pyrolysis data of butane isomers [34] were also used to validate minor C2 + C2 pathways, i.e. C₂H₅ + C₂H₅ and C₂H₅ + C₂H₄. The performance of the present model is shown in Figs. S21–S26 in the *Supplementary Materials*. The sub-mechanisms of C₄H₆, C₄H₄ and C₄H₂ were further validated against the previous pyrolysis data of C₄H₆ isomers, C₄H₄ and C₄H₂ in the literature [80–84], as shown in Figs. S35–S44 in the *Supplementary Materials*. Generally, the present model can capture the literature results well compared with the other three models [27,44,51].

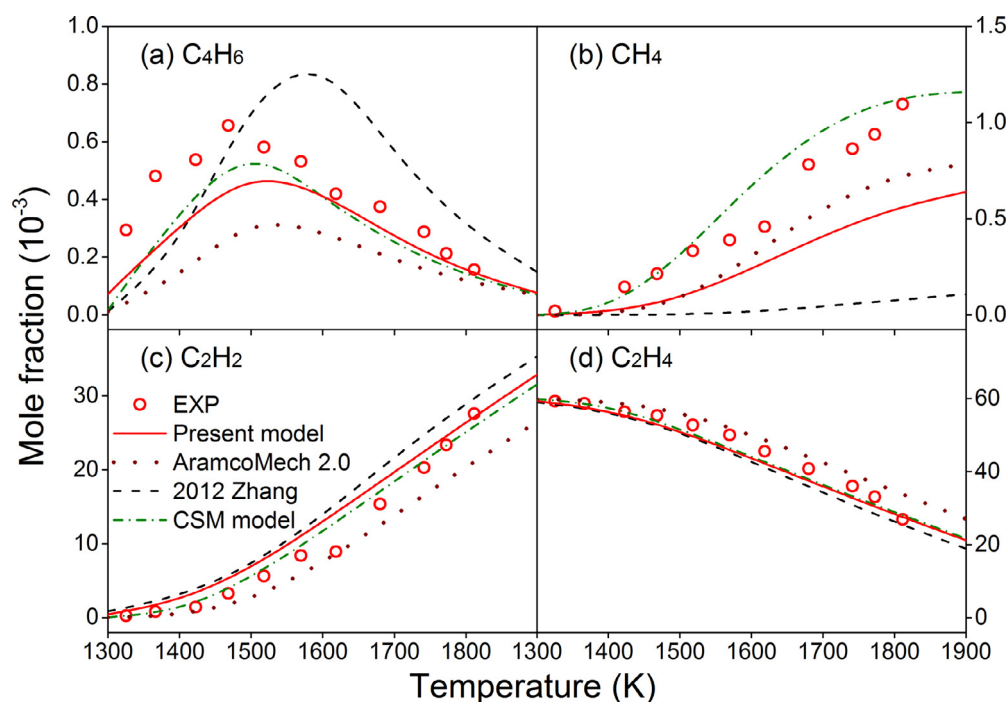


Fig. 8. Comparison between the measured [76] (symbols) and predicted (lines) results in the shock tube pyrolysis of C_2H_4 at 2 bar. The mixture is 6% C_2H_4 /94% Ar. The effective heating times are 1.93 ms (1300 K), 1.81 ms (1400 K), 1.7 ms (1500 K), 1.59 ms (1600 K), 1.47 ms (1700 K), 1.36 ms (1800 K), 1.24 ms (1900 K), respectively. Solid, dash dot, dotted, dashed lines represent the predicted results of present model, CSM model [44], AramcoMech 2.0 model [51] and 2012 Zhang model [27], respectively.

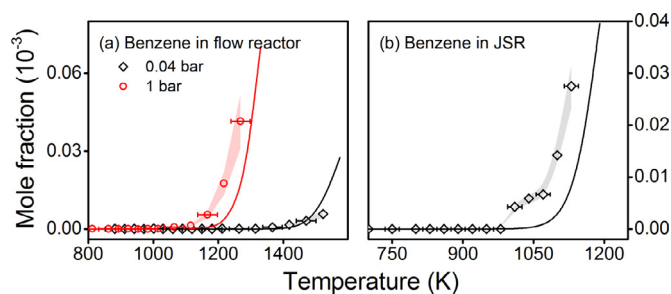


Fig. 9. The measured (symbols) and predicted (lines) mole fraction profiles of (a) benzene in the flow reactor pyrolysis of butane-2,3-dione at 0.04 and 1 bar; (b) benzene in the JSR pyrolysis of butane-2,3-dione at 10 bar. The mole fraction and temperature uncertainties are shown with shadows and bars, respectively.

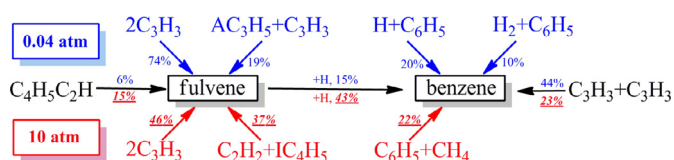


Fig. 10. ROP analysis of benzene formation at 0.04 bar (1523 K) and 10 bar (1130 K), where the C_2H_6 conversion reaches $\sim 80\%$. Numbers without and with underlining represent the contributions at 0.04 and 10 bar, respectively.

4.4. Formation of benzene

The measured and predicted results for benzene in the pyrolysis of butane-2,3-dione are shown in Fig. 9. The present model can well capture the formation of benzene at 0.04 bar, while it underpredicts its formation at atmospheric and high pressures. The ROP analyses were conducted at 1523, 1269 and 1130 K for the 0.04, 1 and 10 bar pyrolysis of butane-2,3-dione, respectively, where the C_2H_6 conversion reaches $\sim 80\%$, as shown in Fig. 10. The most important pathway leading to the formation of benzene at 0.04 bar is

R33, i.e. the self-combination of propargyl radical (C_3H_3). In contrast, the H-assisted isomerization of fulvene (R39) contributes the most to the formation of benzene at 1 and 10 bar. Besides, the H abstraction reaction of phenyl radical to form benzene also has increasing importance as the pressure increases. Although the recombination reaction of propargyl radical (R34) is the most important reaction leading to the formation of fulvene at all investigated pressures, this pathway contributes more to the formation of fulvene at 0.04 bar. The other $C_3 + C_3$ pathway, i.e. R35, is only important to the formation of fulvene at low pressure. At atmospheric and high pressures, the combination reaction of 1,3-butadien-2-yl radical (IC_4H_5) and C_2H_2 (R38) becomes another important source for fulvene formation.

In general, $C_3 + C_3$ pathways are dominant to the formation of benzene and fulvene at low pressure, while these pathways become less important at atmospheric and high pressures. In contrast, the H abstraction reaction by phenyl radical, $C_4 + C_2$ pathways and fulvene isomerization play more and more important roles in benzene and fulvene formation as the pressure increases. This reveals that larger precursors of benzene and fulvene could be more favorably produced at high pressure. Take C4 precursors as an example, under the investigated conditions, C4 species are mainly originated from the combination of $CH_3 + AC_3H_5$ to produce 1- C_4H_8 , which is a pressure-dependent reaction. Increasing the pressure greatly promotes the formation of 1- C_4H_8 . Therefore, the subsequent C4 species like C_4H_5 and C_4H_3 radicals are more favorably formed at high pressure, which are important precursors leading to the formation of fulvene and phenyl radical, respectively. Besides, under the investigated conditions, unimolecular reactions are more favorable at low pressure while bimolecular reactions at high pressure. For C4 species, take IC_4H_5 as an example, unimolecular decomposition reaction is predominant at 0.04 bar while its predominant role was replaced by the bimolecular reaction like R38 ($IC_4H_5 + C_2H_2 = \text{fulvene} + H$) at 10 bar. Consequently, the $C_4 + C_2$ pathways are more important for the formation of fulvene at high pressure, which readily isomerizes

to benzene via H-assisted reaction. Additional efforts should focus on the pressure-dependent rate constants of R33–35, R38 and R39 or other reaction channels leading to the formation of benzene in order to better predict the measured mole fractions of benzene. Besides benzene, a signal at $m/z=92$ which corresponds to toluene was detected only at 1 bar in the flow reactor with peak mole fraction ~ 4.7 ppm. Due to the low concentration of toluene, the formation reactions of toluene are not incorporated in the present model. No signals larger than $m/z=92$ were detected in both the flow reactor and JSR experiments.

5. Conclusion

The pyrolysis of butane-2,3-dione was performed in a flow reactor at 0.04 and 1 bar over 780–1520 K, and also conducted in a JSR at 10 bar between 650 and 1130 K. Reaction intermediates were measured using several diagnostic methods, including SVUV-PIMS, FTIR, GC and GC/MS. A pyrolysis model of butane-2,3-dione, incorporating the sub-mechanisms of butane-2,3-dione, CH_2CO , C1–C4 hydrocarbons and the formation of benzene, was constructed in this work. It was validated against the present data and pyrolysis data of CH_2CO and C2–C4 hydrocarbon fuels in the literature. Some conclusions can be drawn:

- For the decomposition of butane-2,3-dione, the unimolecular decomposition reaction R3 plays a dominant role at low pressure over the whole investigated temperatures. In contrast, at atmospheric and high pressures, R3 and R5 (H abstraction reaction of butane-2,3-dione by CH_3) compete with each other to consume the fuel. At low butane-2,3-dione conversion, R5 has larger contribution while R3 becomes more important at high fuel conversion.
- Besides CH_2CO and CH_3COCH_3 , which have been identified as early oxygenated products of butane-2,3-dione, CH_3CHO was also observed and measured in the present flow reactor and JSR experiments. It is also an early product in butane-2,3-dione pyrolysis and some possible formation pathways were discussed in this work. The formation of these oxygenated species is found to be influenced by pressure.
- The growth pathways from CH_3 to benzene are also dependent on pressure. At 0.04 bar, addition-elimination reactions play an important role while their contributions become less at atmospheric pressure. At 10 bar, the combination reactions instead of addition-elimination reactions are responsible for the carbon chain growth. For the formation of benzene and fulvene at 0.04 bar, $\text{C}_3 + \text{C}_3$ pathways are the main source while their contribution drops at 1 and 10 bar. H abstraction reaction by phenyl radical, $\text{C}_4 + \text{C}_2$ pathways and fulvene isomerization reaction also contribute to the formation of benzene or fulvene at atmospheric and high pressures.
- Finally, the present experimental data provide additional validation targets for C1–C4 pyrolysis model, especially covering the intermediate temperature range (800–1200 K) which has received little attention in the literature. The present pyrolysis model for C1–C4 hydrocarbon fuels could reasonably predict the present data and selected validation targets in the literature.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2018.11.003](https://doi.org/10.1016/j.combustflame.2018.11.003).

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